

- 6 (a) Explain what is meant by the term 'root-mean-square value' as applied to an alternating current.

.....
 [1]

- (b) An alternating current varies with time as shown in Fig. 6.1.

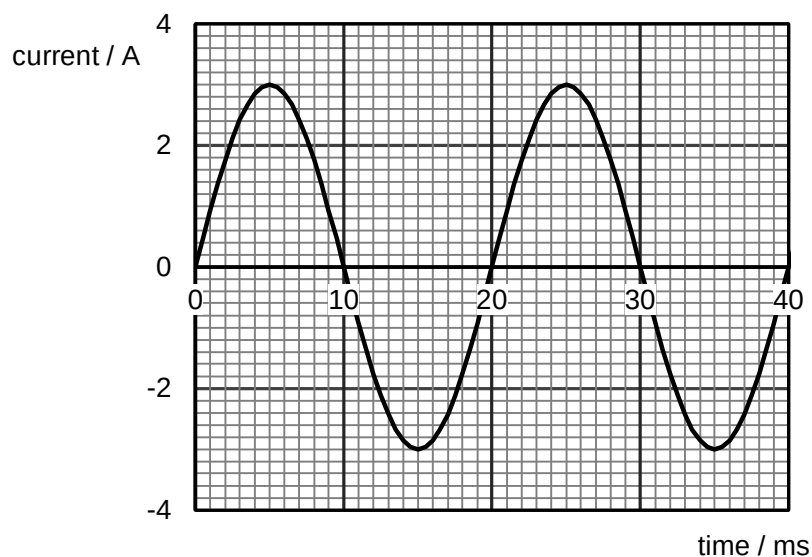


Fig. 6.1

Use the graph to determine, for this alternating current,

- (i) the frequency,

frequency = Hz [1]

- (ii) the peak value,

peak value = A [1]

- (iii) the root-mean-square value.

root-mean-square value = A [1]

- (c) On Fig. 6.2, sketch a graph which shows how the power supplied by this current to a resistor of resistance $5.0\ \Omega$ varies with time from 0 to 40 ms. Mark on the vertical axis the maximum value of the power.

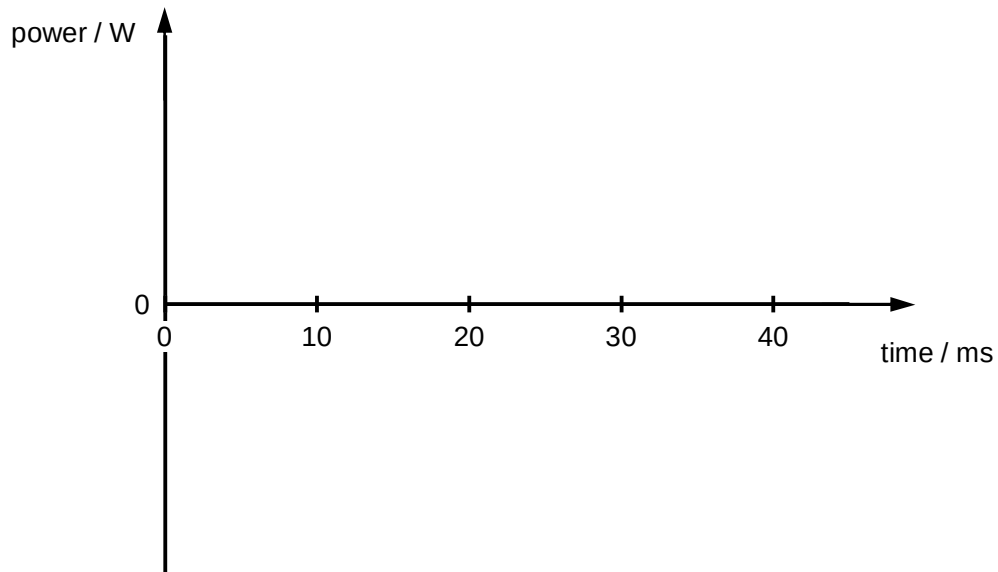


Fig. 6.2

[2]

- (d) The current shown in Fig. 6.1 is in the 300-turn primary coil of an ideal transformer. The secondary coil of the transformer has 6000 turns.

Calculate the transformer's peak output current.

peak output current = A [2]

7 (a) (i) State what is meant by a photon